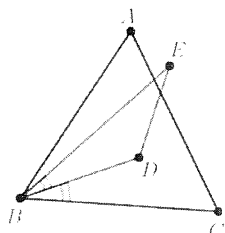


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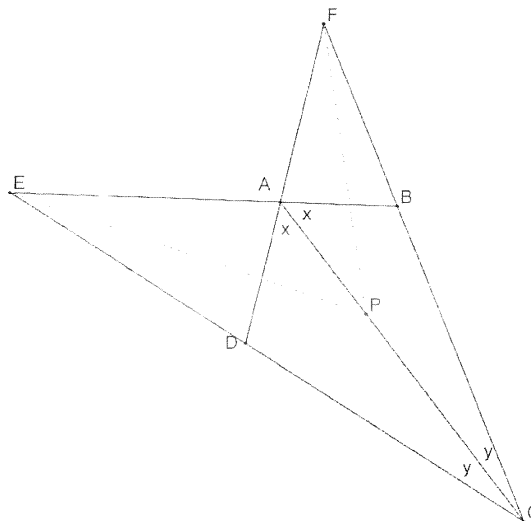
SUB-JUNIOR 2020 -2021

Answer ALL Questions

1. $\triangle ABC$ is an equilateral triangle. D is a point inside $\triangle ABC$ such that $AD = BD$. Choose E such that $BE = AB$ and BD is a bisector of $\angle CBE$. Find $\angle BED$. Diagram given below.



2. Given a quadrilateral $ABCD$, the diagonal AC bisects both $\angle A$ and $\angle C$. If AB and DC extended intersect at E , and AD and BC extended intersect at F , show that for any point P on line AC , $PE = PF$.



3. Twenty-seven balls labelled from 1 to 27 are distributed in three bowls red, blue, and yellow. What are the possible values of the number of balls in the red bowl, if the average of the labels in the red, blue, and yellow bowl are 15, 3, and 18 respectively?
4. Prove that the integer $53 \times 83 \times 109 + 40 \times 66 \times 96$ is not a prime number.
5. Let a, b, c , and d be real numbers such that $a^3 + b^3 + c^3 + d^3 = a + b + c + d = 0$. Prove that sum of a pair of these numbers is equal to 0.
6. Find any hundred digit number with *non-zero* digits, that is divisible by the sum of its digits. Explain the process by which you arrived the answer.
7. $\frac{a}{b}$ is an improper fraction in its simplest form. If its equivalent mixed fraction is $d\frac{c}{b}$, such that $d + c = 9$ and $a + b = 100$, then find all such fractions.
8. $ABCDEFGH$ is a regular heptagon. How many quadrilaterals can be obtained from the heptagon satisfying the following properties:
- All the four vertices of the quadrilateral are some vertices of the heptagon
 - Two sides of the quadrilateral are also sides of the heptagon, but not the other two.